RESEARCH PAPER

Class-agnostic candidate generation and calendar-derived event windows for rapid landslide mapping

Luoyang Xiong[a] and Juan Meng[a]

[a]University of Emergency Management, Sanhe, Hebei 065201, China

ARTICLE HISTORY
Compiled October 28, 2026

ABSTRACT
Triggered landslides are mapped months after the event, whereas response needs an inventory within days and no labelled example exists when candidates must be produced. Two decisions set how many landslides reach a map: which route proposes candidates, and which images are differenced for change channels. Across eleven triggering events, each held out, a class-agnostic proposer placed 2.13 to 3.11 times more landslides into a fixed number of candidates than models that must name what they find, and the gap widens further when only 100 candidates per image are kept. A training-free statistic from the event's change signal tracked the winner (rank correlation 0.661, correlation 0.915, ten of ten signs correct). The pre- and post-event dates follow from the event date and the image dates, reproducing the annotated pair exactly in 97.93% of 3,135 images and within one in all of them. Committing the pair in training raised recall by 5.4% at no cost at inference, an event-level gain that a leave-one-region-out fold does not reproduce. The results show that identification should be designed for candidate generation rather than recognition, at about 300 candidates per image, and the pre-event/post-event pair belongs with the sensor specification, not with the tuning of the model. On the four test events, the proposer places 71 of every 100 mapped landslides among the first 900 candidates it returns. Against the months a full inventory takes, one event yields within a day of the post-event image a candidate set that supports warning and response.

KEYWORDS
landslide inventory; rapid post-event mapping; Sentinel-1 and Sentinel-2; change detection; landslide hazard; disaster response; event-disjoint evaluation

1. Introduction

Landslides are among the most destructive natural hazards worldwide, and they are mapped under a deadline that the imagery does not respect: an inventory is needed within days of the triggering event, yet it is usually assembled months later, because annotation can only begin once the imagery exists and manual interpretation has taken place (Hoehn et al. 2025).

Landslide inventories are the working material of hazard assessment and risk planning, and their quality propagates into every model built on them (Guzzetti 1999; Loche et al. 2022; Lima et al. 2021). What has changed is the delivery side: satellite constellations now image a triggering event within days, and remote sensing is

CONTACT Juan Meng. Email: mengjuan@cidp.edu.cn

already the operative route for detection, monitoring and prediction (Casagli et al. 2023), with forecasting effort concentrated on the predictability of individual slopes (Intrieri and Gigli 2016). Reported case studies show what that means in practice: inventories for major rainfall events are still assembled after the fact and analysed with rainfall and runout models rather than produced during the response window (Jain et al. 2021), and the global record of fatal landslides continues to grow (Fidan et al. 2024). The bottleneck this paper addresses is therefore not the model but the first step of the workflow: turning a freshly acquired image into a ranked set of candidate landslides that manual interpretation can screen. Sentinel-1 and Sentinel-2 now deliver free, global and day-and-night observations at a cadence far beyond what manual interpretation can cover, so the operational question is no longer whether imagery is available but whether an automated system can place the right candidates where manual interpretation can act on them quickly enough to matter (Ghorbanzadeh 2022; Fan 2025).

Supervised detection, and its dependence on the event it saw. Most automated landslide mapping is formulated as supervised recognition: a detector or a segmentation network is trained on annotated landslides and evaluated on held-out images of the same events. Newly triggered failures, however, rarely carry annotations at the moment they must be mapped, and models trained on one event are routinely evaluated on images drawn from the same inventory, so a reported accuracy describes performance inside an event rather than transfer to the next one (Wei et al. 2026; Ullo et al. 2021; Liu et al. 2021). What cross-event transfer does not preserve is the appearance statistics of the annotated events, a sensitivity already visible when segmentation models are transferred between regions (Ghorbanzadeh et al. 2021).

Foundation-model adaptation, and its dependence on a description. A second and newer line replaces the trained head with an adapted visual foundation model. Adapters anchored on frozen backbones have reduced cross-domain error at the pixel and candidate-body level, and segmentation frameworks built on boundary enhancement or on prompt-driven foundation models now report gains on target areas with few labels (Liu et al. 2026; He et al. 2026; Li et al. 2026; Fu et al. 2026), and cross-regional few-shot segmentation has been benchmarked on a multi-region corpus (Fan et al. 2026). These studies establish two premises this paper adopts (that transferability rather than in-domain accuracy determines operational value, and that held-out units must be isolated) but they evaluate accuracy on the segmentation output. What they do not examine is the stage that precedes segmentation: whether the true landslides enter the candidate set at all. A failure that never appears as a candidate cannot be recovered by a better adapter.

Susceptibility modelling, and a lesson this paper carries over. A third family ranks terrain rather than localising individual failures, and it has produced the most systematic evidence about how evaluations of a natural-hazard map behave. Within that family, a single data-pipeline decision has repeatedly been shown to move the reported map further than the choice of model. The way non-landslide samples are drawn shifts the map, and how far it shifts depends on the learner applied to it, with deep models proving less sensitive to the sampling strategy than statistical ones (Dou et al. 2023, 2020). The grid resolution at which conditioning factors are sampled changes both which factors are ranked essential and how much they move the map (Zhao et al. 2024; Liao et al. 2022). The mapping unit changes the prediction itself, because slope units and grid cells partition the same terrain differently (Chang et al. 2023). The order in which hybrid models are integrated is another such decision (Yang et al. 2026), and when such choices are varied jointly the estimates turn out

less robust than their reported accuracies suggest (Ozturk et al. 2021). Whole-model comparisons within a single region belong to the same genre, whether they contrast statistical, hybrid and physically based approaches or compare model families under one protocol (Ferreira et al. 2025; Bui et al. 2022; Azarafza et al. 2021), and the same logic has been used to standardise an entire task family (Vivone et al. 2021), as it has for the second large landslide benchmark (Xu et al. 2024).

The two decisions that are still left implicit. Although this literature has examined how a model is built, two decisions that sit in the candidate-discovery stage are still typically made without being reported. The first is which family performs discovery: a detector trained on labelled landslides, a text-conditioned detector that localises a described class, or a class-agnostic proposer that localises regions without naming them. The second is which two acquisitions are differenced to build the change channels. The default processing of the released data takes the first and last acquisition of an image's series as its pair, a span of well over a year (called the long baseline in this paper), whereas the annotated event window is a single adjacent pair inside that span; the same baseline is then applied to unannotated images, because an annotation-derived window does not exist for them. Comparisons across discovery families are further reported without the candidate budget that each arm produces (the number of candidates kept per image), so a ratio quoted between two arms can reflect how many candidates each exported rather than how well each localised (Hoehn et al. 2025). Both decisions are framed by the distribution-shift literature as properties of the trained model (Peng et al. 2022; Luo et al. 2025), which is precisely why they are not reported as specification items.

This study. To address these two gaps, this study fixes the event window with a rule derived from the acquisition calendar, compares five discovery arms (each arm is one configuration of the discovery stage) at a reported candidate budget under event- and region-disjoint partitions, and asks where in the pipeline the window must be applied. Three questions are addressed in order.

(1) Does the choice of discovery route set the number of landslides that enter the candidate pool on events and regions that were never seen? Five arms covering three paradigms are scored through one export path and one scorer, with the actual candidate count of every arm reported, so that a ratio can be read at a cap every arm reaches.

(2) Can the annotated event window be recovered without any annotation? The event window is treated as a property of the acquisition dates: bracketing the event date with the image's own acquisition series is tested against the annotated window over 3,135 images from eleven inventories, and its tolerance to an imprecise date and to missing acquisitions is measured rather than assumed.

(3) At which stage must that window be applied: training or deployment? The two choices are crossed, and the consequence of applying the correction in the wrong place is measured. The size of the gain is bound to the partition on which a team intends to deploy, so it is reported for the event-level partition used here rather than as an average across partitions.

The answers are useful to practice in one further respect. Because the window is an operation on the acquisition dates rather than a learned one, and because the candidate budget of every arm is reported, a mapping team can decide before training whether the discovery route matters for the event in hand, and how many candidates per image the team must screen to reach most of the attainable recall.

2. Study events and data

The unit of this study is the triggering event, not a slope or a catchment, and that choice decides both how the data are organised and which quantity scores a result. A single study area would hold one terrain and one trigger fixed; the claim tested here is transfer to an event the model has not seen, so the partitions are drawn over events and over regions, and the eleven events occupy the place a study-area subsection would take in a susceptibility study.

For the same reason a result is scored by how many mapped landslides enter a bounded candidate list (AR@k) rather than by the segmentation accuracy of a held-out tile. Accuracy measured inside one event answers a question this paper does not ask, and it would be high for every arm compared here, including the two that place fewer than a fifth of the mapped landslides into the candidate pool; what distinguishes the arms is whether the landslides reach the list at all, and that is what AR@k counts.

2.1. Events and inventories

Table 1 lists the eleven triggering events of the Sen12Landslides benchmark used in this study (Hoehn et al. 2025), and Figure 1 places them. The events span six single-event inventories (rainfall and flooding in Emilia-Romagna, Cyclone Idai in Chimanimani, the June 2018 rainfall in south-western Honshu, the Mj 6.7 Iburi earthquake in Hokkaido, Typhoon Mangkhut in Itogon, and the January 2020 earthquake sequence in Puerto Rico) together with five multi-event inventories from China, Indonesia, Kyrgyzstan and New Zealand. The south-western Honshu inventory is dated 28 June 2018 and its annotated pair runs from 26 June to 1 July, that is, before the rainfall episode of 5–8 July 2018 that struck the same region, so the inventory should not be read as covering that later event. Together they supply 4,392 annotated images and 49,484 bounding boxes at a native 128×128 footprint on a 10 m grid.

The inventories differ in the way that matters for a transfer test: lithology ranges from karst and loess to volcanic terrain and the inventories were collected under contrasting climatic settings (Hoehn et al. 2025), the trigger alternates between rainfall, cyclone and earthquake, and the annotated objects differ in size by more than a factor of three (median side 3.9 px in Emilia-Romagna against 13.2 px in the Chinese inventory). Two inventories span several years and several triggers, so their acquisition dates are per image rather than per inventory; for those, the window used here is the highest-confidence candidate returned by the dating record of the released data, a choice fixed before any comparison was run.

Read as a set, the eleven events span Europe, Africa, Asia, Oceania and the Caribbean, carry three trigger types (rainfall and flooding, cyclone, earthquake), cover both inventory granularities (a single dated event and a multi-year national catalogue), and differ threefold in the size of the annotated object. That spread is what makes the transfer test informative, and it is the reason every comparison below is reported per fold rather than as one pooled number.

Terrain differs as much as the triggers. Over the 4,392 annotated tiles the median elevation of a tile runs from 38 m in Puerto Rico to 2,098 m in the second Kyrgyz inventory, and the 98th percentile of elevation inside a tile reaches 3,166 m in the Chinese inventory; the median slope of a tile ranges from 7.2° to 40.5°, with nine of the eleven events between 16.7° and 33.0°. Both quantities are computed from the elevation channel of the released DEM, per event, with the script archived alongside

Table 1. Eleven Sen12Landslides events used in this study. Inventory type and source span follow the released dataset; the event date is the one assigned by its NDVI-based dating module, and the trigger is the documented mechanism. Tiles and boxes are counted from the released manifest (4,392 annotated tiles, 49,484 boxes in total); *dated tiles* are those for which the acquisition calendar yields a pre/post pair (3,135 in total). The magnitude of the Hokkaido event is on the Japan Meteorological Agency scale (Mj 6.7); its moment magnitude is Mww 6.6 in the USGS catalogue.

Event	Country / region	Type	Event date	Trigger	Split	Tiles	Boxes	Median side (px)
italy	Italy (Emilia-Romagna)	Single	2023-05-16	rainfall and flooding	train	1,799	35,293	3.9
chimanimani	Zimbabwe (Chimanimani)	Single	2019-03-15	Cyclone Idai	test	345	2,285	7.7
hiroshima	Japan (SW Honshu)	Single	2018-06-28	heavy rainfall	train	322	2,011	7.4
hokkaido	Japan (Iburi, Hokkaido)	Single	2018-09-06	Mj 6.7 earthquake	val	144	2,400	9.5
itogon	Philippines (Benguet, Luzon)	Single	2018-09-15	Typhoon Mangkhut	train	82	313	6.9
china	China	Multi	2016-11-21 – 2021-09-21	multiple	test	47	92	13.2
indonesia	Indonesia	Multi	2016-05-15 – 2019-09-22	multiple	test	253	1,714	8.4
kyrgyzstan1	Kyrgyzstan	Multi	2016-04-01 – 2019-09-11	multiple	train	694	2,247	11.0
kyrgyzstan2	Kyrgyzstan	Multi	2017-01-29 – 2019-08-07	multiple	train	170	469	8.8
newzealand	New Zealand	Multi	2016-06-18 – 2022-02-07	multiple	val	508	2,524	12.0
usa	USA (Puerto Rico)	Multi	2020-01-07	earthquake sequence	test	28	136	5.5
Total	11 events					4,392	49,484	

the code; the annotated objects track the same spread, from a median side of 3.9 px in Emilia-Romagna to 13.2 px in the Chinese inventory (Table 1). Table S1-L of the supplementary material lists the two summaries per event.

2.2. Imagery, modalities and the change channels

Four input modalities are used for every event, and a tile is a 15-channel tensor: pre-event Sentinel-2 surface reflectance (3 channels); the ascending Sentinel-1 synthetic aperture radar (SAR) stack (4 channels, VV and VH, pre- and post-event); topography from a digital elevation model (DEM) (5 channels: elevation, slope, aspect, curvature and a topographic wetness index); and the event-change representation (3 channels, ΔNDVI, ΔNDWI and ΔNBR). Which two acquisitions are differenced is the first of the two decisions examined here, and Section 3.2 shows that it can be fixed from the acquisition dates alone.

All arms operate on a common 320×320 input obtained by resampling the native grid identically for every modality. The resampling keeps the comparison fair but adds no spatial detail, and it is the reason the smallest annotated objects remain only a few pixels across: candidate recall is bounded by the pixel size of the target, not by the capacity of the model. Normalisation statistics are computed on the training split only.

2.3. Partitions

Three partitions are used, and no event appears in more than one subset of a given partition, so every reported test event is absent from training. The event-level partition assigns five events to training (Emilia-Romagna, both Kyrgyzstan groups, Hiroshima and Itogon), two to validation (Hokkaido and New Zealand) and four to test (Chimanimani, China, Indonesia and Puerto Rico). Leave-one-event-out uses seven folds, each holding out one event entirely. Leave-one-region-out uses three folds (East Asia, Central Asia and South-East Asia), each holding out a whole region that is absent from training, which makes it the partition closest to deployment. Fold definitions are released with the manuscript.

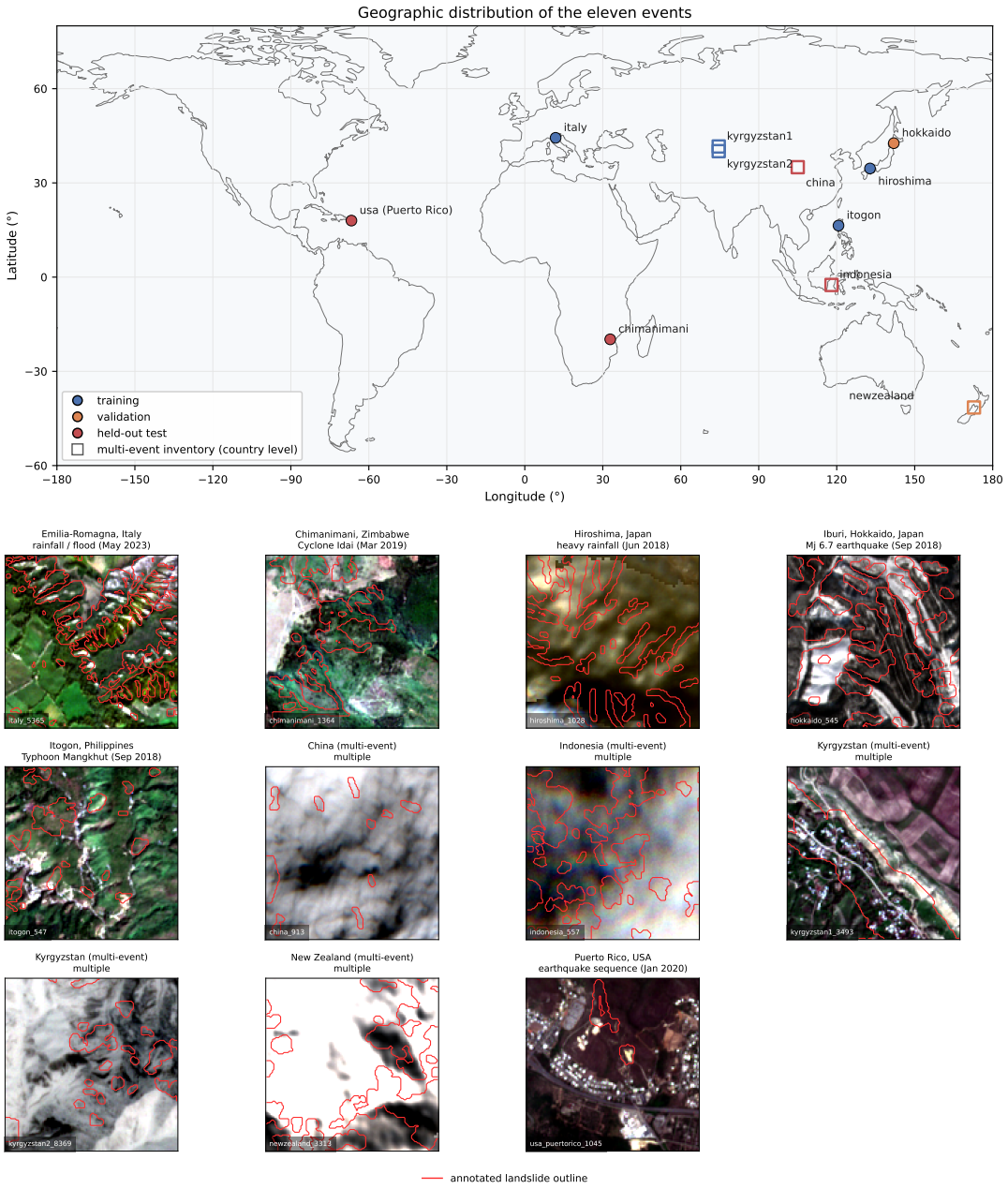


Figure 1. Geographic distribution of the eleven events (top) and one representative Sentinel-2 pre-event image per event with the annotated landslide outlines (bottom). Filled markers are single-event inventories; open squares are multi-event inventories drawn at country level.

3. Method

3.1. Overview of the pipeline and of what is compared

Figure 2 shows the pipeline studied here. It has three stages, and the paper's two research questions are attached to the first two. Candidate discovery turns an image into a ranked set of proposals; the event-window construction step decides which two acquisitions are differenced to build the change channels; verification reduces the can-

didate set to a mapped inventory and is held fixed throughout, so that differences observed at discovery are not absorbed by the verifier.

Two choices inside this pipeline are usually left implicit. The first is which family of detector performs discovery: a closed-set detector trained on landslide labels, a text-conditioned detector that localises a described class, or a class-agnostic proposer that localises regions without naming them. The second is which pair of acquisitions defines the change channels. This study fixes the second choice with a rule derived from the acquisition calendar (Section 3.2), holds verification fixed, and varies the first across five arms at a reported candidate budget (Section 3.3). All arms are scored through one export path and one paradigm-neutral scorer (Section 3.4).

The eleven events of Table 1 supply 4,392 annotated images and 49,484 boxes at a native 128 × 128 footprint on a 10 m grid. Every arm operates on the same 320 × 320 input, obtained by upsampling the native grid identically for all modalities; this keeps the comparison fair but adds no spatial detail, so the smallest annotated objects remain only a few pixels across (Section 4.2).

3.2. The Event-Bracketing Rule

The change channels of an image are built by differencing a pre-event and a post-event acquisition. The long baseline takes the first and last acquisition of the series as its pair, a mean span of 678 days, while the annotated window is a single adjacent pair located near the middle of the series. **Event-Bracketing Rule (EBR)**: given the event date *and nothing else*, take the last acquisition strictly before the date and the first acquisition strictly after it, and use that adjacent pair as the change window.

The rule operates on acquisition dates, not on learned features. It needs no imagery, no annotation and no training, and, because it reads only dates, it can be applied to images that carry no annotation at all, which is the situation of the negative images used during training.

How closely does it reproduce the annotated window? Over 3,135 images from the eleven inventories for which the dating record returns a single date, EBR reproduces the annotated pre/post indices exactly in **97.93%** of images and within one acquisition in **100.00%**, with no image for which a single event date fails to bracket it. Per inventory the exact rate ranges from 0.759 (Kyrgyzstan, second group) to 1.000 (Italy, Hiroshima, Hokkaido, Itogon, Puerto Rico); the four inventories below 0.93 are exactly the multi-event ones, whose annotated windows are not tied to a single date. The event sits a median of 3 days after the previous acquisition and 4 days before the next, with a 90th percentile of 7 days on both sides, which is the physical reason the rule works: the series is dense enough that the two frames adjacent to an event are almost always the ones manual interpretation would choose.

Robustness to an imperfect event date. A published event date is rarely exact, so the rule was re-evaluated with the date perturbed by δ days (Table 3). A one-day error costs almost nothing (0.948 for δ = -1, 0.970 for δ = +1) and the pair remains adjacent in 100% of images. A three-day error costs exactness of the pair (the exact-match rate falls to 0.260 for δ = -3, 0.488 for δ = +3) but the pair is still adjacent within one acquisition in **99.5%** of images, so the window stays on the correct side of the event. The failure is asymmetric: an early date breaks the bracket faster than a late one, because moving the date before an acquisition turns that acquisition into the post-event frame, whereas moving it later mostly reuses the same pair. Beyond one week the rule collapses (0.073 at δ = -7, 0.047 at δ = +7); the bound is worth stating,

because a mis-dated inventory is the one input the rule cannot survive.

Robustness to missing acquisitions. Randomly withholding acquisitions from the acquisition calendar gives the second failure mode (Table 3, lower block). Dropping 2% of acquisitions reduces the exact rate to 0.842, 5% to 0.669 and 10% to 0.433, while the within-one-acquisition rate degrades much more slowly (0.989, 0.943 and 0.816). Missing acquisitions are therefore more damaging than a wrong date of the same nominal size: losing 5% of the series costs about as much of the exact-match rate as a two-day date error, and the rule degrades gracefully in adjacency even when it loses the exact-match rate. For operational use this implies a concrete check (verify that the image's series carries the expected cadence before applying the rule) rather than a statement that the rule is robust to data gaps.

Scope. A further 1,252 images (28.5% of the 4,392 annotated images) carry not one but a set of two to eleven candidate dates, and are excluded from the validation above because there is no single date to bracket. For these the pipeline uses the highest-confidence candidate, a deterministic rule the dataset supports, and **no exact-match rate figure is claimed for them**. The rule is therefore established on the 71.5% of images whose dating is unambiguous, and the paper reports that restriction rather than averaging over it.

3.3. Candidate-discovery arms and their budgets

Five arms cover three paradigms (Table 2). **A class-agnostic proposer** is the adopted route of this study: a prompt-free universal region proposal network that localises object-like regions without category supervision (Tang et al. 2026), fine-tuned on the training events with the four input modalities; the design descends from class-agnostic proposal networks (Ren et al. 2017) and from localisation networks trained without classification (Kim et al. 2022). **Two fine-tuned detectors** represent supervised recognition: a closed-set detector and a text-conditioned open-vocabulary detector (Liu et al. 2024; Cheng et al. 2024), both trained on the same five training events. **Two zero-shot detectors** represent the transfer-without-retraining assumption of the open-vocabulary literature, applied at their default configurations (Liu et al. 2024; Cheng et al. 2024).

The candidate counts are not equal by construction, and that is reported rather than adjusted away. At a common cap of 900 proposals per image the actual median count is 900 for the proposer and for Grounding DINO on every fold, 359–900 for fine-tuned YOLO-World (it fills the cap on one of the ten folds), 744–900 for zero-shot YOLO-World, and 118–296 for the closed-set detector. Every arm's actual count is reported per fold in the supplementary material, and the paradigm comparison is additionally read at the strict common cap of 100 proposals per image, at which every arm can be matched. This distinction matters for the interpretation: a ratio computed at 900 candidates is a comparison of what each arm reaches, whereas a ratio computed at 100 is a ratio at a cap every arm reaches.

3.4. Scoring, metrics and protocols

Metric. Candidate discovery is scored by average recall at k proposals per image, AR@k for $k \in 100, 300, 900$, at an intersection-over-union threshold of 0.5. AR@k is the mean of per-image recall over the images that contain ground truth, so images without ground truth are excluded rather than counted as zero. It measures the recall

ceiling, that is, the fraction of annotated landslides that reach a fixed number of candidates per image. Read as a count, a value of 0.71 means that 71 of every 100 mapped landslides are among the first 900 candidates an analyst is shown, which is the quantity the downstream verification stage inherits. The ground-truth-weighted counterpart (micro) is reported alongside every arm, because the two differ by 0.03–0.05 AR on identical proposals (0.3321 image-averaged against 0.2900 weighted for one fine-tuned arm at 900 candidates). Unless a value is explicitly labelled weighted, every comparison in this study is image-averaged.

Why not a false-positive-rate metric. The adapted proposer assigns scores in a narrow range (approximately 0.31–0.51), so recall at a low false-positive rate is uninformative at the discovery stage. This is a property of the proposal stage rather than a limitation of the evaluation, and it is reported so that the metric choice is transparent. Not reporting a false-positive rate, an ROC curve or an overall classification accuracy is therefore a design decision rather than an omission: none of the three is defined for an arm that emits a ranked candidate list without class decisions, which is exactly the stage this paper compares.

Paradigm-neutral scoring. Arms with incomparable native scores are made rankable by exporting every arm's proposals and scoring them with one shared scorer. The scorer uses no class labels in the main comparison, so that the observed differences cannot be attributed to how a score is interpreted; a version with semantic content is reported separately (Section 4.5).

Protocols. Three partitions are used, and no event appears in more than one subset of a given partition, so every reported test event is absent from training. The event-level partition assigns five events to training, two to validation and four to test. Leave-one-event-out (LOEO) uses seven folds, each holding out one event entirely. Leave-one-region-out (LORO) uses three folds, each holding out a whole region that is absent from training; because no event straddles subsets, LORO is the protocol closest to deployment. Fold definitions are released with the manuscript.

3.5. Implementation and reproducibility

All arms receive the same four modalities (Sentinel-2 pre-event reflectance, the ascending Sentinel-1 stack, DEM-derived topography and the event-change channels) and the same 320×320 input. Normalisation statistics are computed on the training split only. The proposer is trained for eight epochs; the detectors are fine-tuned on the same five training events with their released configurations.

Inference cost measured on a single consumer GPU is 37.2 ms and 3.8 GB of peak memory per image for the proposer, against 19.7 ms and 1.3 GB for zero-shot YOLO-World and 203.2 ms and 2.5 GB for zero-shot Grounding DINO. The comparison is therefore not one of equal computational cost, and the cost of each arm is reported with its result.

The pipeline is not numerically deterministic: re-running one held-out fold end to end reproduces the archived value to 0.0013 AR, and the run-to-run variation across folds and seeds is 2.2–2.9 points of AR@900. The paradigm and modality contrasts exceed that variation by at least eight times; the remaining contrasts (the training-window gain and its partition-specific variants) are smaller than the unpaired variation and are therefore adjudicated by paired same-seed replicates rather than by their margin over it. Data, fold definitions, the bracketing rule and the scoring scripts are archived in a public repository whose link is given in the data availability statement.

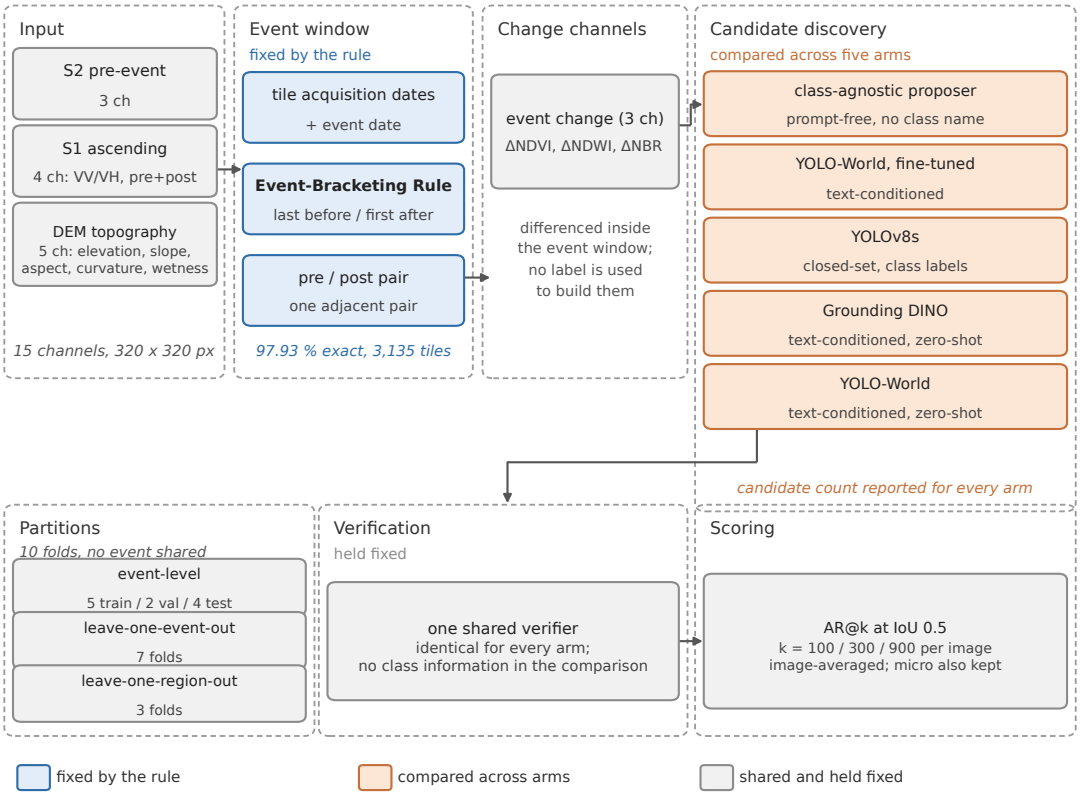


Figure 2. Pipeline and the two decisions studied. The tile is 15 channels at 320×320 px (3 optical, 4 SAR, 5 topography, 3 change). Blue marks the decision fixed by the rule: the event window follows from the acquisition dates and the event date, and it reproduces the annotated pair exactly in 97.93% of the 3,135 single-date tiles. Orange marks the decision compared across five arms at a candidate count reported for every arm; the change channels are differenced inside the window chosen by the rule and no label is used to build them. Verification is shared and held fixed for every arm (grey), and every arm is scored by AR@k at an IoU of 0.5 under the three partitions.

Table 2. The five discovery arms. The last column is the actual median number of candidates per image across the ten cross-setting folds; it is not a design parameter, and it is reported so that every ratio in Section 4 can be read against the count the arm produced.

Arm	Discovery paradigm	Conditioning	Realised median candidates
Class-agnostic proposer (this study)	class-agnostic localisation	none	900 (every fold)
YOLOv8s	closed-set recognition	class labels	118–296
YOLO-World (fine-tuned)	open-vocabulary recognition	class description	359–900
Grounding DINO	open-vocabulary recognition	class description	900 (every fold)
YOLO-World (zero-shot)	open-vocabulary recognition	class description	752–900

Table 3. Event-Bracketing Rule under two failure modes. Upper block: the event date is displaced by δ days with the acquisition series fixed. Lower block: a fraction of the acquisitions is withheld at random. Reference: the annotated pre/post pair of 3,135 images. *Exact* counts a pair reproducing both indices, *within one* a pair adjacent to the annotated one.

δ (days)	−3	−2	−1	+0	+1	+2	+3
exact	0.260	0.828	0.948	0.979	0.970	0.607	0.488
within one acquisition	1.000	1.000	1.000	1.000	1.000	0.999	0.995
unbracketable	1	0	0	0	0	1	10
acquisitions withheld	0%	2%	5%	10%	20%		
exact	0.979	0.842	0.669	0.433	0.179		
within one acquisition	1.000	0.989	0.943	0.816	0.513		

4. Results

The results are organised so that the evaluation setup comes first and the two transferable findings follow. Section 4.1 reports what the three discovery paradigms retain on held-out events and regions; Section 4.2 asks whether that outcome could have been forecast without training; Section 4.3 takes the event window in turn and locates the pipeline stage at which it must be applied; Section 4.4 converts the recall curve into a number of candidates an operator can plan with, and Section 4.5 asks what the remaining pool is worth once a verifier is attached.

4.1. How many landslides reach the review list depends on the discovery route

On the held-out events. All five arms were scored through one export path and one scorer on the four test events, none of which appears in training (Figure 4, Table 2). The results separate into three tiers. Read as counts per 100 mapped landslides, the proposer places 71 of them among the first 900 candidates an analyst is shown, against 33 for the strongest fine-tuned arm and 18 for the best zero-shot one. The two models applied without target-domain training retain little: Grounding DINO reaches AR@900 = 0.1828 and YOLO-World applied zero-shot 0.1019; the latter returned no proposal at its default threshold and assigned a maximum confidence of 3.6×10^{-4} to any region of this imagery, so its failure is a property of the model on this data rather than of a conservative threshold. The two models fine-tuned on the five training events recover to about half the recall of the class-agnostic proposer: 0.3321 for YOLO-World and 0.3203 for YOLOv8s. **The class-agnostic proposer reaches 0.7059 at the same 900-candidate cap, and 0.6619 when the same proposals are scored ground-truth weighted.** On this split the two fine-tuned arms land on the same level although one conditions on labels and the other on text, so what separates them from the proposer here is the discovery paradigm rather than the conditioning. The strongest external arm at this cap is not one of the fine-tuned models but Grounding DINO applied zero-shot, which reaches a ten-fold mean of 0.2524 against 0.2222 for fine-tuned YOLO-World and 0.1674 for zero-shot YOLO-World: a model applied to these events without any training on target-domain labels still returns a better candidate pool than one trained on them. That ordering is the sharpest form of the argument in this paper. Target-domain supervision, which is the expensive resource in this literature (Casagli et al. 2023), does not on its own provide a better candidate pool; what it provides is the class decision, which is precisely what a class-agnostic route defers. Figure 3 shows what that difference looks like on a single tile: one of the proposer's 900 candidates lands on the reference object at IoU 0.67, while the five candidates the closed-set detector returns for the same tile miss it entirely.

The ratio depends on the cap at which it is read, and that dependence, not the ratio itself, is what the comparison reports. At the cap of 900 candidates the proposer stands 2.13-fold above the fine-tuned text-conditioned arm and 3.86-fold above the untrained one. At the strict budget of 100 candidates per image (a budget every arm can fill) the same contrast widens to 7.7-fold and 10.1-fold, and the two fine-tuned arms become indistinguishable (0.0617 against 0.0606). The contrast therefore shrinks monotonically as the budget grows.

Across ten cross-setting folds. Figure 4 and Table 4 repeat the comparison under leave-one-event-out and leave-one-region-out partitions, with all ten folds scored

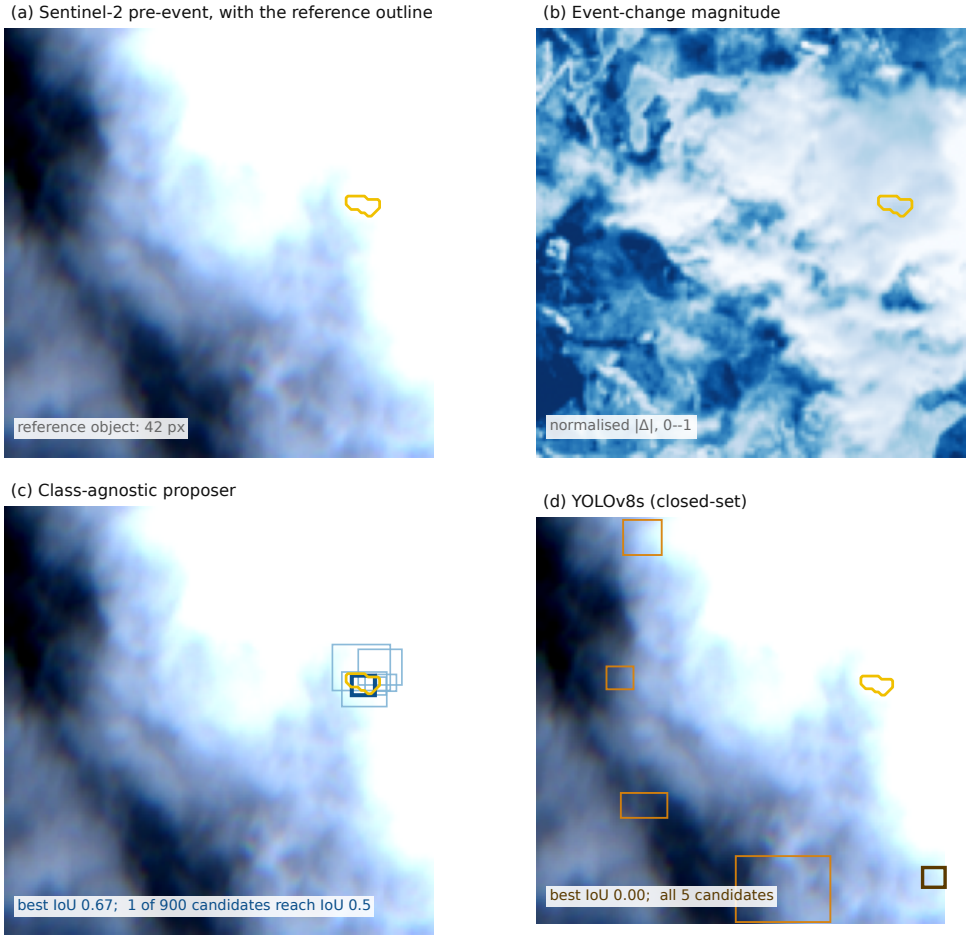


Figure 3. What one candidate looks like, on one tile (`italy_1161`, Emilia-Romagna) shown at its native 128×128 footprint. (a) Pre-event Sentinel-2 image with the reference outline (the largest annotated component of the tile, 42 pixels, about 10×6 pixels). (b) Event-change magnitude, the norm of the change stack. (c) The class-agnostic proposer's best candidate over that outline, IoU 0.67, with its next five candidates by IoU; one of its 900 candidates for this tile reaches IoU 0.5. (d) All five candidates the closed-set YOLOv8s returns for the same tile, none of which overlaps the object (best IoU 0.00). IoU is measured against the bounding box of the reference outline, the convention used for scoring throughout.

through the same path (Figure 4). The proposer reaches a ten-fold mean AR@900 of 0.6903, against 0.2222 for fine-tuned YOLO-World, 0.2524 for Grounding DINO applied zero-shot and 0.1674 for YOLO-World applied zero-shot; YOLOv8s reaches 0.0978 when allowed up to 900 candidates and 0.0433 at the strict common cap of 100. Two ratios must therefore be distinguished at this cap: 3.11-fold against the fine-tuned text-conditioned arm, and 2.73-fold against the strongest external arm on this partition, which is Grounding DINO applied zero-shot. The ten-fold comparison is read at the cap of 900 candidates; the budget dependence of the ratio is reported on the four-event split, where the same arms collapse to 0.0617 and 0.0606 at 100 candidates and the contrast to the proposer widens rather than narrows (Section 4).

Why the candidate count has to be reported. Compared without their candidate budgets, these arms can appear to differ by two orders of magnitude, and that ratio is a property of the exports rather than of the paradigms: when an arm's export is capped below the budget at which the ratio is read, its AR@900 is computed

from a pool it never filled. Table 2 reports the actual median count for every arm on every fold (900 for the proposer and for Grounding DINO on every fold, 359–900 for fine-tuned YOLO-World, 744–900 for YOLO-World applied zero-shot, and 118–296 for YOLOv8s) and with those counts on the page the ratio settles at 2.13-fold and 3.11-fold. **Part of the magnitude reported for this task in earlier work reflects the candidate budget of the arms rather than the paradigm.**

Regions are not what limits the candidate pool. Across the three leave-one-region-out folds the proposer reaches a mean AR@900 of 0.7262 (s.d. 0.0395), which is not lower than the 0.6749 (s.d. 0.1597) reached across the seven leave-one-event-out folds, even though the region-level partition removes an entire region from training. The lowest fold is Italy at 0.2896, the largest inventory and the one with the smallest annotated objects (median side 3.9 px, Table 1); the remaining six folds fall between 0.6907 and 0.7889.

4.2. The choice can be anticipated from the event's own change signal

The question. If the paradigm gap were a property of the event rather than of the model, it should be visible before training. A cheap property of the imagery was therefore tested: the separability of the event's own change signal, computed from the annotated masks and the change channels, with no model in the loop.

The association. Figure 5 plots that statistic against the measured gap for the ten held-out folds. The relation is monotone rather than exactly linear, so both coefficients are reported: a Pearson correlation of 0.915 (95% CI [0.682, 0.986]), a rank correlation of 0.661, a leave-one-fold-out correlation of 0.856 when the relation is refitted on nine folds and applied to the tenth, and the correct sign in ten of ten folds. The rank figure is the relevant one for a statement about ordering, because a single fold with a large gap can carry a linear coefficient.

A two-regime description. The separation is sharp enough to describe the ten folds as two groups rather than a continuum. Where separability exceeds 0.70 Grounding DINO, the weaker route on every fold, recovers 17.7–26.9% of the proposer's recall (gap 3.71–5.65; 22% of the proposer's recall at the group mean), and where it falls below 0.50 it recovers 63.7–72.8% (gap 1.37–1.57; 67% at the group mean); no fold lies between 0.492 and 0.705 on that axis.

What the statistic is and is not. It is a decision aid, not an automatic rule. It presupposes a reference inventory for the event, because the statistic is computed from existing masks, and the 0.50 and 0.70 cuts describe these ten folds rather than a calibrated threshold. Within that scope it forecasts which route will win and roughly by how much on an event whose landing objects are already mapped, it says nothing about a newly triggered event for which no mask exists yet, and it does not predict the AR@900 value itself. Where no reference inventory is available (the situation this paper is designed for) the statistic cannot be computed at all, and the route must be chosen on the systematic evidence of Section 4.1 rather than on a per-event forecast. Its practical role is narrower and still useful: it tells a mapping team, before any model is trained, whether the choice of discovery route is likely to matter much for the event in hand.

4.3. The pre-event/post-event pair must be fixed before training

The design (Figure 6, Table 5). Two decisions that the literature treats as one were separated: which window the model is *trained* on, and which window it is *deployed* with. Table 5 crosses the two choices, holding architecture, data and candidate count fixed; the training window is either the long baseline or the bracketed event pair, and the deployment window is either the same or the other one.

Order matters. The results show that the correction cannot be applied at inference time. A model trained on the long baseline and then given the correct event window at deployment falls from AR@900 = 0.7085 to 0.6217, a loss of 0.087, that is, from 71 to 62 of every 100 mapped landslides placed among the first 900 candidates; it has learned filters for a contrast it is no longer shown. The reverse order works: a model trained on the event window but deployed on the long baseline rises from 0.7094 to **0.7474** over three seeds, a gain of 0.0380 (+5.4% relative), growing to **+11.7% at 100 candidates per image**, at no inference cost and with no architectural change. Against the run-to-run variance of 0.0059 taken from three archived runs of the same configuration, that puts the AR@900 gain at 5.5 standard errors, the AR@300 gain at 13.0 and the AR@100 gain at 8.7. A paired per-image bootstrap over the 673 test images that carry ground truth gives an independent second estimate, +0.0468 at k = 900 with a 95% interval of [+0.029, +0.065].

Why it works. Training on the event pair teaches the change encoder to respond to event-driven change: the widened window is anchored to the event and 98.5% of its pixels are usable in both frames, against 77.6% for a single adjacent pair. At deployment the long baseline supplies a larger contrast (a scar visible at the event is still visible 678 days later) and filters trained this way exploit it better than filters trained on the long baseline itself, which also key on seasonal drift and dilute the event signal.

The change channels carry most of the accuracy. The modality ablation (Table 6) shows where the signal sits: an optical-only configuration reaches AR@900 = 0.3533, adding SAR and topography raises this to 0.4574, and adding the event-change channels raises it to 0.7167; across three seeds the means without and with the change channels are 0.4642 and 0.7127, a difference of +0.2485 that survives paired testing (q = 0.00075).

The gain is tied to the partition. Holding out the whole Indonesian region and retraining on the other ten inventories, committing the window in training lowers AR@900 from 0.7891 to 0.7717 and AR@300 from 0.7154 to 0.6731 (paired over 253 images: -0.0173, 95% interval [-0.041, +0.006]; -0.0423, [-0.067, -0.018]), and at k = 100 the loss is 0.0535 with an interval that excludes zero. The comparison is tight because the event-level partition scores the same 253 images, where committing the window gains 0.0414. **What carries across partitions is the window itself (recoverable from the acquisition calendar and unrecoverable after training) while the size of the gain is bound to the partition on which a team intends to deploy.**

4.4. What a candidate budget provides a mapping team

The curve. Figure 7 turns the pool into a resource decision. On the event-level test split the proposer reaches AR@900 = 0.7059, but the marginal return per additional candidate falls from about 5.5×10^{-3} between 10 and 30 candidates to 1.2×10^{-4} between 500 and 900. At 300 candidates the route already attains 0.5908, that is 84% of what it reaches at 900, for one third of the candidates.

The cost. Inference costs 37.2 ms and 3.8 GB of peak memory per image for the proposer, against 19.7 ms and 1.3 GB for YOLO-World applied zero-shot and 203.2 ms and 2.5 GB for Grounding DINO, which reaches 0.1828. The comparison is therefore not one of equal computational cost, and every arm's cost is reported with its result.

The operating point. Taken together, these results support a concrete operating point: **about 300 proposals per image**, at which the pool holds 84% of the attainable recall for a third of the candidates. What that pool is worth downstream depends on verification, which is held fixed here and reported in Section 4.5.

4.5. What the pool is worth once a verifier is attached

The stage. Discovery produces candidates; verification decides which of them become a mapped inventory. Because the verification stage is shared and paradigm-neutral here (Section 3.4), the results at that stage measure how much of the candidate pool can be recovered, not the performance of a deployable inventory.

How much survives ten proposals per image. With the verifiers tested in this study, ten proposals per image recover 1.9–2.9% of the annotated landslides, that is, fewer than three of every 100, and a vision-language verifier (CLIP) with a selection balanced across arms lifts that to 3.3% (Figure 8); the proposer's own verifier, scoring its own proposals, reaches 22.7% at the same cap, so the gap between what enters the pool and what survives verification is an order of magnitude. This number is not an inventory result: at ten candidates per image it is a screening quantity, and the operating point recommended in Section 4.4 (300 proposals per image) is the one at which a mapping team would be screening rather than mapping.

How much of the paradigm gap survives. The results show that a verifier which uses no class information removes most of the separation observed at discovery: at the strict cap of ten proposals per image, Grounding DINO comes within about 5% of the proposer's yield. Once the verifier uses class information, the separation returns to 1.8–2.6-fold. The contrast between paradigms is therefore not an artefact of the verifier, but its magnitude does depend on what the verifier can use (a property of the verification stage rather than of the discovery stage, and one that a study reporting discovery alone would miss).

The reading. Taken with Sections 5.1 and 5.4, these results place the limit at discovery and the remaining uncertainty at verification: the choice of discovery route changes how many landslides enter the candidate pool by a factor of two to three, while the verifier determines how much of that pool becomes usable.

5. Discussion

5.1. Why a route that does not name the class transfers better

The first result is that recall on held-out events separates the three paradigms, and the separation has a common cause in each family. A detector trained on labelled landslides internalises the appearance statistics of the events it saw and has no mechanism for the statistics of the next one (Wei et al. 2026); a text-conditioned detector must instead match image content to a description, and it is the description (not the geometry of the failure) that has to transfer across events (He et al. 2026). A class-agnostic proposer avoids both because its objective is to place a region around anything locally object-like and to defer the semantic decision; the advantage observed here therefore

Table 4. Per-fold average recall at 900 candidates per image on the ten cross-setting folds (image-averaged, IoU 0.5). The second column gives the number of annotated boxes scored in each evaluation partition; the two partition families (leave-one-event-out and leave-one-region-out) are alternative partitions of the same imagery, so their counts are not additive. The actual candidate count of every arm is given in Table 2; s.d. is the between-fold standard deviation, which is not the standard error of a paired difference quoted in the text; the comparison at the strict common cap of 100 candidates is reported in the text.

Fold	GT boxes	Class-agnostic	YOLO-World (ft)	YOLOv8s	Grounding DINO	YOLO-World (zs)
Italy (LOEO)	35,293	0.2896	0.1051	0.0350	0.0515	0.0337
Kyrgyzstan 1 (LOEO)	2,247	0.7559	0.2694	0.1399	0.4835	0.3441
New Zealand (LOEO)	2,524	0.6907	0.4430	0.1604	0.5031	0.3166
Chimanimani (LOEO)	2,285	0.7350	0.1540	0.0532	0.1550	0.0920
Hiroshima (LOEO)	2,011	0.7315	0.1779	0.0737	0.1481	0.0963
Indonesia (LOEO)	1,714	0.7889	0.2502	0.0908	0.1675	0.0772
Hokkaido (LOEO)	2,400	0.7330	0.2105	0.0972	0.1981	0.1832
East Asia (LORO)	4,503	0.7307	0.2001	0.0877	0.1937	0.1405
Central Asia (LORO)	2,716	0.6757	0.2487	0.1207	0.4443	0.3055
South-East Asia (LORO)	2,027	0.7721	0.1625	0.1192	0.1795	0.0845
Mean	—	**0.6903**	0.2222	0.0978	0.2524	0.1674
s.d.	—	0.1373	0.0877	0.0365	0.1526	0.1082

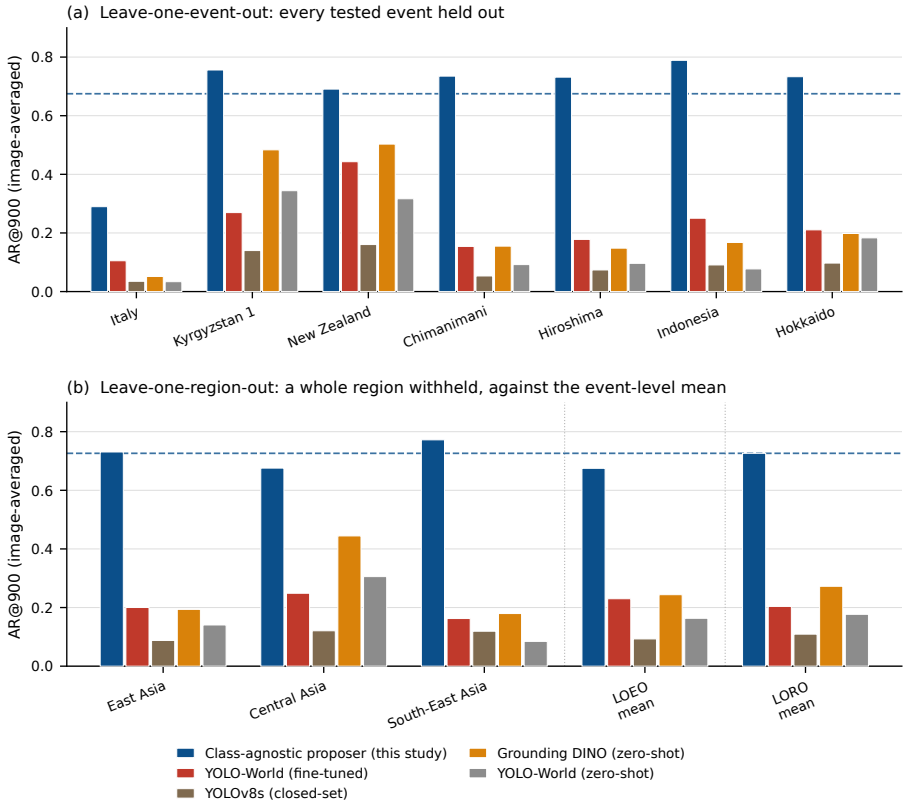


Figure 4. Cross-event recall (a) and cross-region transfer (b) for the ten cross-setting folds, plotted from the values given in Table 4. All five discovery arms are shown at 900 candidates per image, image-averaged; dashed lines mark the partition mean of the class-agnostic proposer (0.6749 over the seven event-level folds and 0.7262 over the three region-level folds). The last two groups of panel (b) repeat the two partition means for every arm.

comes not from localising landslides better but from **not requiring a decision that shift makes unreliable**.

The mechanism is testable, and one alternative is excluded. If the proposer's advantage rested on the change channels rather than on the paradigm, re-

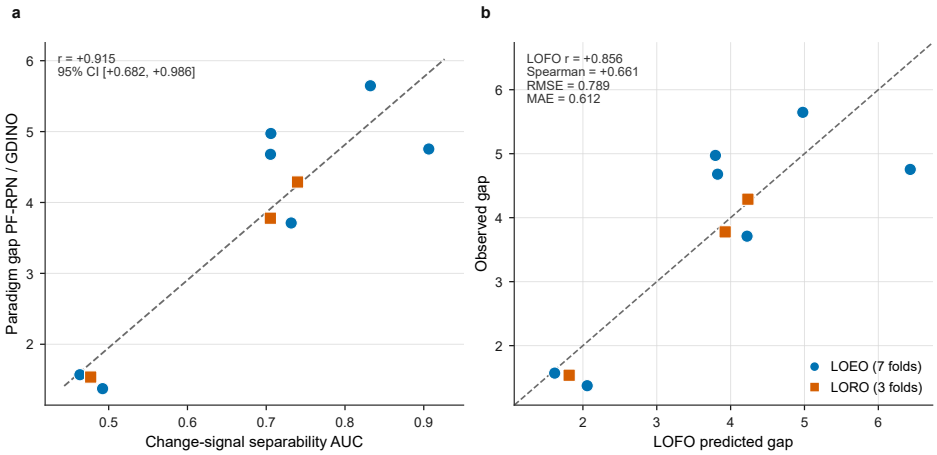


Figure 5. Separability of the event change signal against the measured paradigm gap on the ten folds (left) and the leave-one-fold-out prediction of that gap (right).

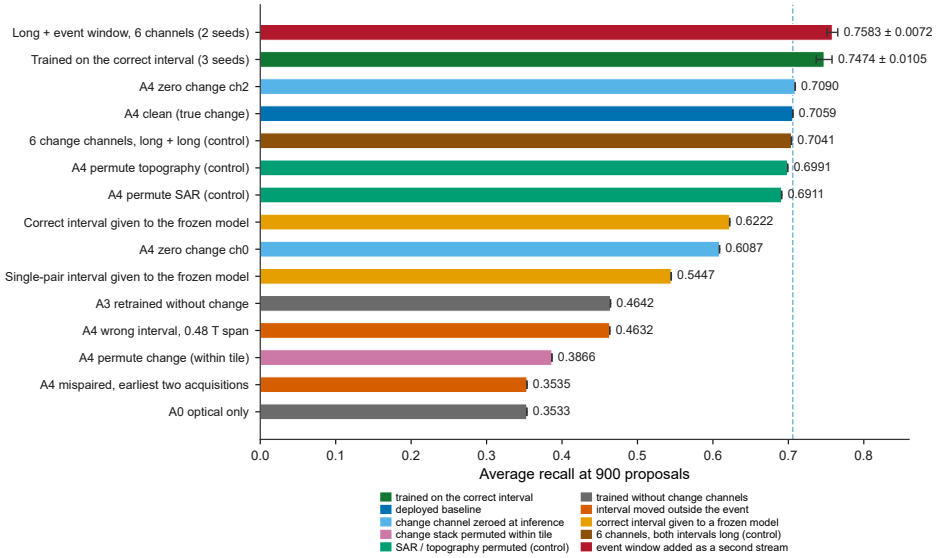


Figure 6. The event interval changes recall only if it is committed during training. AR@900 on the event-level test split, sorted by value; the same arms are tabulated in Table 5.

Table 5. The two interval choices crossed. Training window against deployment window, with architecture, data and candidate budget held fixed. The two frozen rows replace only the change stack of a frozen anchor model evaluated on a second machine whose own anchor is 0.7085, so their difference is computed within that block.

Training window	Deployment window	AR@900	Difference
long baseline	long baseline	0.7094	anchor
long baseline, stack duplicated to 6 ch	long baseline (6 ch)	0.7041	−0.005
single adjacent pair	long baseline	0.6969	−0.013
widened event window (3 seeds)	long baseline	**0.7474**	+0.038
long + widened event (6 ch, 2 seeds)	long + widened (6 ch)	0.7583	+0.049
single adjacent pair	single adjacent pair	0.5783	−0.131
widened event window (1 seed)	widened event window	0.7179	+0.009
long baseline (frozen)	widened event window	0.6217	−0.087

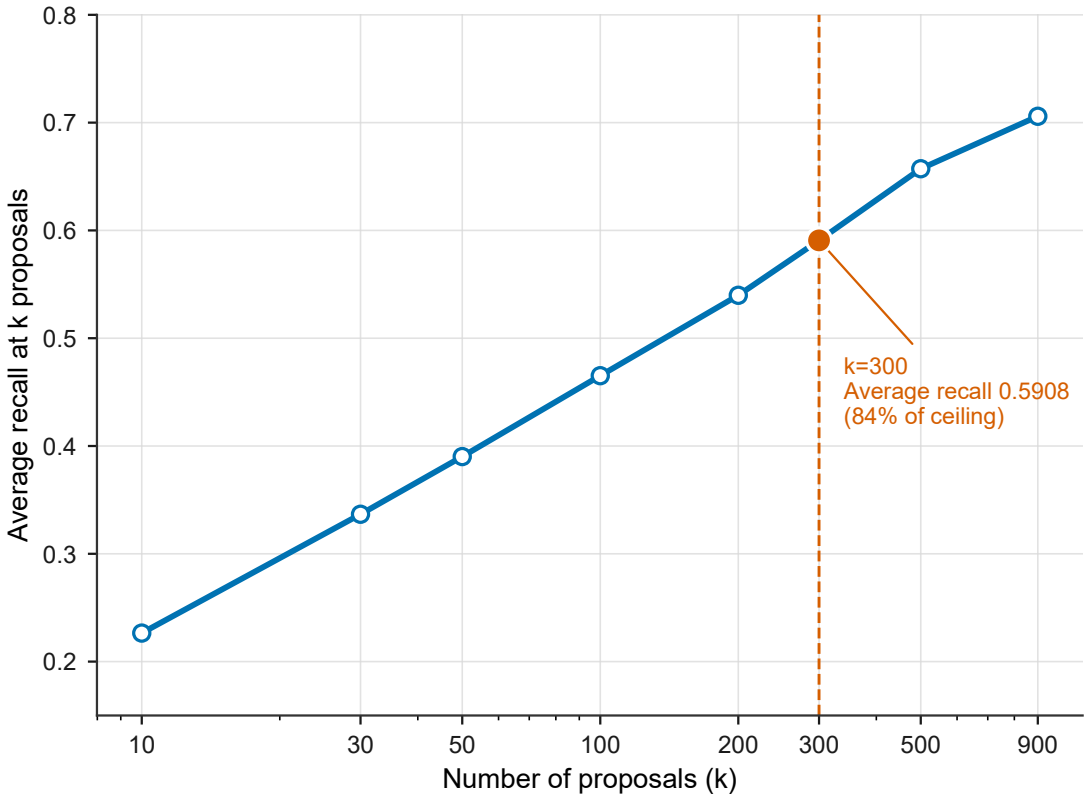


Figure 7. Average recall as a function of the number of candidates per image on the event-level test split; the annotated point marks the recommended operating point at 300 candidates.

Table 6. Modality ablation on the event-level split (A0–A9): average recall at 100, 300 and 900 candidates per image. Values are transcribed from the archived ablation run (1,759 images, 4,227 ground-truth boxes); the A4 configuration of that run scores 0.7167 here against 0.7059 for the checkpoint used in Section 4, a difference of 0.0108 that is reported rather than averaged.

Configuration	AR@100	AR@300	AR@900
RGB baseline	0.0962	0.1974	0.3533
Optical + DEM	0.1426	0.2527	0.4220
Optical + SAR	0.1439	0.2622	0.3745
Optical + SAR + DEM	0.1590	0.2878	0.4574
A3 + temporal change	0.4882	0.6171	0.7167
A4 + terrain-aware token	0.4678	0.5859	0.6905
A5 + cross-modal router	0.4797	0.6000	0.7059
A6 w/o router	0.4797	0.5998	0.7029
A6 w/o SAR	0.4712	0.5858	0.6729
A6 w/o change	0.1652	0.3048	0.4799

moving those channels should remove the advantage. It does not: with optical, SAR and topography but no temporal input the route still reaches AR@900 = 0.4642 across three seeds, against 0.7127 with the change channels, and an optical-only configuration reaches 0.3533. The comparison is therefore between paradigms on the same footing, not between one strong and several handicapped arms.

What the result does not establish. The reading above is an interpretation of the pattern rather than a measured mechanism. A competing account (that the proposer is simply more robust to low signal-to-noise input than a model that must commit to a class) is not excluded by these experiments, and is stated here as an alternative rather than dismissed. The statistic itself belongs to the family of transferability

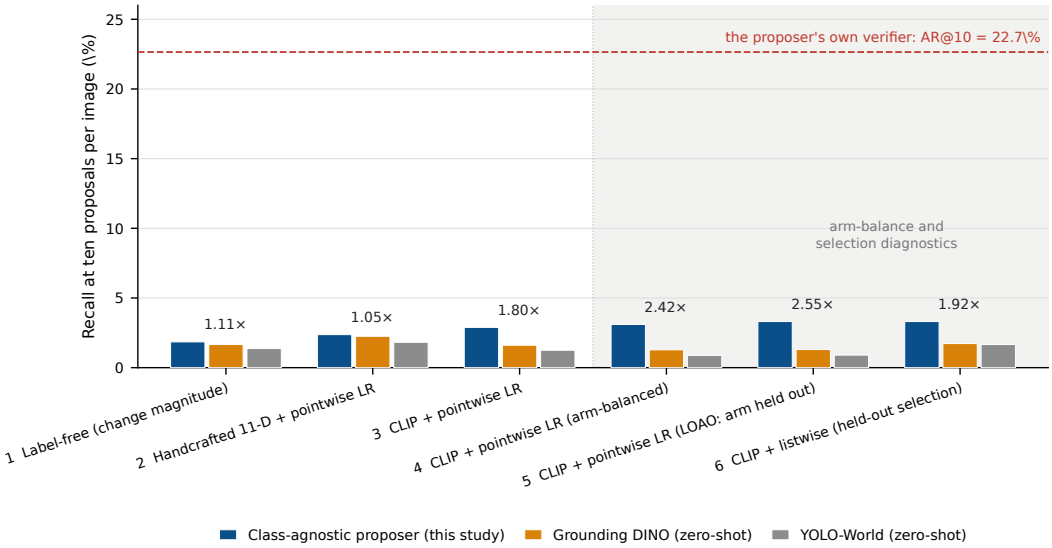


Figure 8. Shared-verifier ablation at ten candidates per image on the event-level test split. Groups 1–3 are the three verifier families tested; groups 4–6 (shaded) are arm-balance and selection diagnostics. The dashed line is what the proposer's own verifier reaches when it scores its own proposals (AR@10 = 22.7%), that is, how much of the candidate pool ten candidates would require; the number above each group is the separation between the class-agnostic proposer and Grounding DINO at the same verifier.

measures that predict target performance without training on the target (Nguyen et al. 2020; You et al. 2021; Pándy et al. 2022). At a matched candidate budget and under event- and region-disjoint partitions, the class-agnostic proposer places two to three times more landslides into the candidate pool than either recognition family.

5.2. The binding constraint is the size of the target at native resolution

The second result concerns not which route wins but where in the pipeline that limit is set. Three observations point the same way. First, removing a whole region from training does not lower recall: the three leave-one-region-out folds average 0.7262, above the 0.6749 of the seven leave-one-event-out folds. Second, the one fold that collapses (Italy, AR@900 = 0.2896) is also the inventory whose annotated objects are smallest at the native grid (a median side of 3.9 px against 7.4–13.2 px elsewhere), and it is also the largest inventory, with 1,799 images and 35,293 boxes, so its result is not a small-sample artefact but the opposite. Third, the candidate-count curve shows that additional candidates return progressively less: recall rises steeply up to about 30 candidates per image and then flattens, with the marginal return per candidate falling from 5.5×10^{-3} to 1.2×10^{-4} between 500 and 900 candidates.

Read together, these observations locate the binding constraint in the pixel size of the target at the native grid rather than in the geographic region. This is consistent with the susceptibility literature in which the resolution at which conditioning factors are sampled changes the reported map more than the choice of model does (Zhao et al. 2024), and it is in line with the data-scarcity argument made for rapid mapping in mountainous terrain (Wei et al. 2026); comparative evaluations of mapping approaches within a single region reach the same conclusion, namely that a pipeline decision can outweigh the choice of model (Ferreira et al. 2025; Bui et al.

2022). It is also the reason the operational recommendation in Section 5.4 is phrased in candidates per image rather than in square kilometres.

That reading rests on two facts: the ordering of LOEO and LORO means and the coincidence of the weakest fold with the smallest objects, and the relation between median object size and recall is not monotone across the remaining six folds; it is therefore offered as a description of this benchmark rather than as a law, and no rule is derived from it.

5.3. Scope of the claims

The claims above are bounded in the following respects, each stated once here.

The gain from the training window is partition-specific. On the event-level partition, committing the window during training while leaving the deployed long baseline unchanged raises AR@900 by 0.0380 (+5.4%, +11.7% at 100 candidates, 5.5 standard errors of the paired difference over three seeds). On a leave-one-region-out fold in which the whole Indonesian region is held out, the same intervention **loses** 0.0173 at AR@900 and 0.0423 at AR@300. The comparison is tight because both partitions score the same 253 images, so what separates them is the backing model rather than the region, and a comparable sensitivity to the way a partition is drawn has been reported when sampling strategies are varied (Dou et al. 2023). What carries across both partitions is the interval itself: it can be recovered from the acquisition calendar, and it cannot be corrected after training; the same statements hold on the region-level fold as well. The size of the gain is the part that must be verified on the partition a team intends to deploy, and the paper reports it that way instead of reporting an average.

The discovery comparison is established within one benchmark. The events span eleven triggering events and several regions, but one sensor pair and one annotation protocol, both inherited from the released dataset (Hoehn et al. 2025). Conclusions about precision at a usable operating point are not supported, and the verification stage is held fixed, so its absolute level (ten proposals per image recover 1.9–2.9% of the annotated landslides) measures how much of the candidate pool can be recovered rather than the performance of a deployable inventory.

Although the pipeline reproduces its inputs exactly, the training is not deterministic. A complete re-run of one held-out fold reproduces the archived AR@900 to 0.0013, and adding two seeds to each of three folds gives fold-wise deviations of 0.0076–0.0251 against a between-fold standard deviation of 0.137; the run-to-run variation is 2.2–2.9 points of AR@900. The paradigm and modality contrasts clear that variation by 8.6–17.0 times. The training-window gain is smaller than the unpaired run-to-run variation, so it is established by paired same-seed replicates (standard errors of the paired difference 5.5, 13.0 and 8.7 at the three caps) and by a bootstrap interval of $[+0.029, +0.065]$ over 673 tiles that excludes zero.

Two further limits apply to the bracketing rule specifically. It tolerates a one-day error in the event date without losing adjacency (the exact-match rate falls only to 0.948 one day early and 0.970 one day late, with the pair still adjacent in every image), and a three-day error costs exactness but not adjacency; the rule also degrades faster under missing acquisitions than under a mis-dated event. The exact-match rate figure itself is established on the 3,135 images whose dating record returns a single date, not on the 1,252 images that carry several candidate dates.

The covariates are the ones the model consumes. The trigger of each event

is recorded in Table 1; the rainfall or seismic intensity behind it belongs to the source inventories and is not reproduced here, and no statement in this paper depends on it. Land cover is not an input. The variables the model actually consumes are characterised quantitatively in Section 2.1, which reports the elevation and slope ranges of every event because they are input channels.

Two different records are involved. The event date of every single-event inventory in Table 1 was checked against records outside the released dataset: the responsible national agency for each event, and the United States Geological Survey catalogue for the two earthquake-triggered inventories, where the local date is the one that matters (Hokkaido is 18:07 UTC on 5 September 2018, as the dataset records, and 6 September in Japan). The dates agree. What the 97.93% figure is measured against, however, is the annotated pre/post pair of the same dataset that supplies those dates, so it establishes that the rule reproduces the annotation record, not that an independent interpreter would have chosen the same pair; separating the two would need an independently produced inventory of one of these events. The five multi-event inventories cannot be checked against a single date at all, because their dates are per image rather than per inventory. Section S1-M of the supplementary material documents the check, the two conventions it depends on, and the procedure for re-running it against an independent date.

5.4. From an event to a review list: the operational workflow

The numbers in this paper can be read as a workflow rather than as a benchmark, and doing so is what makes the word "rapid" concrete. Four terms determine how long it takes to put candidates in front of manual interpretation, and each is measured here rather than assumed.

The delay before the post-event image is set by the acquisition series, not by the method. EBR needs one frame after the event date, and across the 3,135 single-date tiles the event falls a median of 3 days after the preceding acquisition and 4 days before the next (Section 3.2); the post-event frame therefore usually exists by day 4. Nothing in the pipeline can begin before it, and this is the term a mapping team controls least.

The time the proposer takes is negligible at this scale. At 37.2 ms per image on a single consumer GPU the largest event used here (Emilia-Romagna, 1,799 annotated images) is proposed in about a minute, the median event in about ten seconds, and the whole 4,392-image benchmark in under three minutes. Compute is not the cost of the workflow.

The number of candidates is the cost. Ten candidates per image over the whole benchmark is 43,920 candidates and 17,990 for Emilia-Romagna. Review time follows from the rate at which manual interpretation can judge a candidate, which this study does not measure; at one to three seconds per candidate (an interval adopted here as an explicit assumption, not a measurement), that is 12 to 37 hours of review for the benchmark and 5 to 15 hours for the single event, so one event fits inside a day at either end of the interval while the whole benchmark does not. At the per-image ceiling of 300 candidates the same event yields 539,700 candidates, which at the same rates is 150 to 450 hours: not a review task at all, but an annotation task for a machine or for a very large team. This is why the paper separates the two numbers: 300 candidates per image is where the per-image recall curve saturates, and 10 candidates per image is what manual interpretation can finish.

What that implies for a response window. With the frame available at day 4, the proposer finished in about a minute, and a reviewed list of ten candidates per image costing hours rather than days, the deliverable is a ranked list for manual interpretation within a day of the post-event acquisition: not an inventory, and not a statement about map accuracy. The honest comparison is with the months that a full inventory takes, and the honest limit is that the verification stage in this study recovers only 1.9–2.9% of the annotated landslides at that budget (Section 4.5). What the workflow provides is the order and the timing of the work: the calendar rule fixes which frames to difference before any label is spent, the proposer puts candidates on screen within a minute of those frames, and manual interpretation decides where to spend attention.

5.5. What a mapping group should do with this

The findings support a recommendation that is specific about what each number can carry.

Fix the window from the acquisition calendar, before any model is trained. Because bracketing the event date reproduces the annotated window in 97.93% of images without a single label, the window is a specification item (like the sensor and the resolution) and not a tuning knob. It should be documented in the same place as the sensor specification, together with the acquisition series it was derived from.

Design discovery for localisation, not recognition. A class-agnostic proposer places substantially more landslides into a fixed candidate budget than either recognition family, and the difference survives every correction applied here. The practical instruction is to hold the candidate budget equal when comparing routes, and to report the actual count of every arm, because a ratio quoted at unequal budgets can reflect the export cap rather than the paradigm.

300 is where the per-image curve saturates; 10 is what can be reviewed. The two numbers serve different purposes and are not alternatives. At 300 candidates per image the pool already holds 84% of the attainable recall for a third of the candidates, that is, 59 of every 100 mapped landslides for a third of the screening work, and the marginal value of further candidates is small, which makes 300 the point where the per-image recall curve saturates; ten candidates per image is what manual interpretation can finish, and is the number a response plan should be built on (Section 5.4). A statement about the quality of a route is made at 300 candidates per image, and a statement about the review work is made at 10; quoting one in place of the other misstates the work. Inference at 37.2 ms and 3.8 GB per image on a single consumer GPU makes a screening pass over an event feasible on modest hardware.

Treat the pool as a screening aid. Ten candidates per image recover 1.9–2.9% of the annotated landslides with the verifiers tested here, so the system is a candidate generator for manual interpretation rather than an inventory generator. That framing is what makes the cost numbers meaningful: the question is how many candidates an analyst must review, not how many pixels are labelled.

What this provides for a response cycle. The numbers set the division of labour between machine and analyst rather than the accuracy of a product. On a single consumer GPU the proposer needs 37.2 ms and 3.8 GB per image, so the largest event used here (Emilia-Romagna, 1,799 annotated images) is proposed in about a minute of compute; the binding constraint is how many candidates manual interpretation can

review, not how long the model takes. That is why the operating point is stated as a number of candidates rather than as a threshold: at 10 proposals per image the pool already holds a ranked list that manual interpretation can finish, and at 300 it holds 84% of what the route reaches at all. A team that must deliver something before an inventory exists can therefore read the trade-off off one curve, and the calendar rule tells it which two frames to difference before any label is spent.

6. Conclusions

This study examined the two decisions that set the size of the candidate pool in automated landslide discovery (which discovery route is used, and which two acquisitions are differenced to build the change channels) under event- and region-disjoint partitions that keep every tested event out of training. Three findings follow.

A class-agnostic, prompt-free proposer placed two to three times more landslides into a fixed candidate pool than detectors that must name what they find, and the contrast widened rather than narrowed when the cap was cut to the strict 100 candidates per image that every arm could fill. A training-free statistic computed from the event's own change signal tracked which route would win, with a rank correlation of 0.661 against the measured gap, a Pearson correlation of 0.915, and the correct sign in ten of ten folds. The pre- and post-event window itself proved to be a property of the acquisition dates rather than an annotation product: bracketing the event date with the image's own acquisition series reproduced the annotated window exactly in 97.93% of 3,135 images and within one acquisition in all of them, and committing that window during training (while leaving the deployed long baseline untouched) raised AR@900 by 5.4% at no inference cost, an effect that a leave-one-region-out partition does not reproduce and whose size is therefore reported as partition-specific.

The wider lesson is that the event window belongs to the specification of the task rather than to the tuning of the model, and that the training and deployment windows are two independent choices. Three extensions follow directly from the scope set out above: turning verification from a fixed stage into a trained one, repeating the comparison under a second sensor pair and annotation protocol, and applying the calendar rule to other event types whose reference data are also defined by a date. None of the three changes the conclusion of this paper; each would show how far it travels.

Reporting them explicitly, together with the candidate budget produced by every arm, would make comparisons in this literature cheaper and directly comparable; it would also let a mapping group decide, before training anything, whether the discovery route it can afford is the one its events require.

Acknowledgement(s)

The authors thank the developers of the Sen12Landslides dataset for releasing the inventories and the imagery.

Disclosure statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This work was supported by the National Key Research and Development Program of China under Grant 2024YFC3017003.

Data availability statement

The code that implements the Event-Bracketing Rule, the five scoring arms, the separability statistic and the paired bootstrap, together with the fold definitions, the per-tile acquisition tables and the per-image results underlying every figure, is openly available under an MIT licence in the repository `https://github.com/xionglovetian/pfrpn-change-interval` and archived at `https://doi.org/10.5281/zenodo.23730396`; a SHA-256 manifest and a verification script are included, so every archived file can be checked before use. The Sen12Landslides imagery and the event inventories are available from the source dataset (Hoehn et al. 2025) at `https://huggingface.co/datasets/paulhoehn/Sen12Landslides`.

Author contributions

Luoyang Xiong: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft. **Juan Meng:** Conceptualization, Supervision, Funding acquisition, Project administration, Resources, Writing – review and editing.

Use of artificial intelligence

An AI assistant (DeepSeek Harness) was used to help draft and edit the manuscript, to write and debug analysis scripts, and to generate figures from the authors' result files. All scientific statements, numerical values and conclusions were checked and are the responsibility of the authors; no AI tool was used to fabricate data or literature.

Ethics statement

Not applicable: this study uses open satellite imagery and published landslide inventories and involves neither human participants nor animal subjects.

References

M. Azarafza, M. Azarafza, H. Akgun, P. M. Atkinson, and R. Derakhshani. Deep learning-based landslide susceptibility mapping. *Sci. Rep.*, 11(1):24112, 2021. `https://doi.org/10.1038/s41598-021-03585-1`.

Q. D. Bui, H. Ha, D. T. Khuc, D. Q. Nguyen, J. von Meding, L. P. Nguyen, and C. Luu. Landslide susceptibility prediction mapping with advanced ensemble models: Son La province, Vietnam. *Natural Hazards*, 116:2283–2309, 2022. `https://doi.org/10.1007/s11069-022-05764-3`.

N. Casagli, E. Intrieri, V. Tofani, G. Gigli, and F. Raspini. Landslide detection, monitoring and prediction with remote-sensing techniques. *Nature Reviews Earth & Environment*, 4 (1):51–64, 2023. https://doi.org/10.1038/s43017-022-00373-x.

Z. Chang, F. Catani, F. Huang, G. Liu, S. R. Meena, J. Huang, and C. Zhou. Landslide susceptibility prediction using slope unit-based machine learning models considering the heterogeneity of conditioning factors. *J. Rock Mech. Geotech. Eng.*, 15(5):1127–1143, 2023. https://doi.org/10.1016/j.jrmge.2022.07.009.

T. Cheng, L. Song, and Y. Ge. YOLO-World: Real-time open-vocabulary object detection. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pages 16901–16911, 2024. https://doi.org/10.1109/CVPR52733.2024.01599.

Hongqiang Dou, Junbin He, Siyi Huang, Wenbin Jian, and Chaoxu Guo. Influences of non-landslide sample selection strategies on landslide susceptibility mapping by machine learning. *Geomatics, Natural Hazards and Risk*, 14(1):2285719, 2023. https://doi.org/10.1080/19475705.2023.2285719.

J. Dou, A. P. Yunus, A. Merghadi, A. Shirzadi, H. Nguyen, Y. Hussain, R. Avtar, Y. Chen, B. T. Pham, H. Yamagishi, et al. Different sampling strategies for predicting landslide susceptibilities are deemed less consequential with deep learning. *Sci. Total Environ.*, 720: 137320, 2020. https://doi.org/10.1016/j.scitotenv.2020.137320.

X. Fan. Deep learning can predict global earthquake-triggered landslides. *Natl. Sci. Rev.*, 12 (7), 2025. https://doi.org/10.1093/nsr/nwaf179.

Y. Fan, H. Zhang, and J. Chen. Cross-regional few-shot landslide segmentation with visual foundation model via frequency decoupling and iterative refinement. *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, 19:21646–21660, 2026. https://doi.org/10.1109/JSTARS.2026.3706738.

Zuleide Ferreira, Bruna Almeida, Ana Cristina Costa, Manoel do Couto Fernandes, and Pedro Cabral. Insights into landslide susceptibility: a comparative evaluation of multi-criteria analysis and machine learning techniques. *Geomatics, Natural Hazards and Risk*, 16(1): 2471019, 2025. https://doi.org/10.1080/19475705.2025.2471019.

S. Fidan, H. Tanyaş, A. Akbaş, L. Lombardo, D. N. Petley, and T. Görüm. Understanding fatal landslides at global scales: a summary of topographic, climatic, and anthropogenic perspectives. *Natural Hazards*, 120(7):6437–6455, 2024.

Z. Fu, F. Wang, and Z. Tu. LSDSAM: Harnessing visual foundation model and enhanced transfer learning toward practical landslide detection in few-shot scenarios. *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, 19:12344–12365, 2026. https://doi.org/10.1109/JSTARS.2026.3678141.

O. Ghorbanzadeh. The outcome of the 2022 Landslide4Sense competition: Advanced landslide detection from multisource satellite imagery. *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, 15:9927–9942, 2022. https://doi.org/10.1109/JSTARS.2022.3220845.

O. Ghorbanzadeh, A. Crivellari, P. Ghamisi, H. Shahabi, and T. Blaschke. A comprehensive transferability evaluation of U-Net and ResU-Net for landslide detection from Sentinel-2 data. *Sci. Rep.*, 11(1), 2021. https://doi.org/10.1038/s41598-021-94190-9.

F. Guzzetti. Landslide hazard evaluation: a review of current techniques and their application in a multi-scale study, central Italy. *Geomorphology*, 31(1–4):181–216, 1999. https://doi.org/10.1016/S0169-555X(99)00078-1.

Xiang He, Xu Wu, Xiangpeng Wang, Xuqing Ren, and Junbo Guo. Cross-domain coseismic landslide segmentation: local boundary enhancement & global pixel contrastive learning. *Geomatics, Natural Hazards and Risk*, 17(1):2623104, 2026. https://doi.org/10.1080/19475705.2026.2623104.

P. Hoehn, K. Heidler, R. Behling, and X. X. Zhu. A spatio-temporal dataset for satellite-based landslide detection. *Sci. Data*, 12(1), 2025. https://doi.org/10.1038/s41597-025-06167-2.

E. Intrieri and G. Gigli. Landslide forecasting and factors influencing predictability. *Natural Hazards and Earth System Sciences*, 16(12):2501–2510, 2016. https://doi.org/10.5194/nhess-16-2501-2016.

N. Jain, T. R. Martha, K. Khanna, P. Roy, and K. Vinod Kumar. Major landslides in Kerala, India, during 2018–2020 period: an analysis using rainfall data and debris flow model. *Landslides*, 18:3629–3645, 2021. https://doi.org/10.1007/s10346-021-01746-x.

D. Kim, T. Y. Lin, A. Angelova, I. S. Kweon, and W. Kuo. Learning open-world object proposals without learning to classify. *IEEE Robot. Autom. Lett.*, 7(2):5453–5460, 2022. https://doi.org/10.1109/LRA.2022.3146922.

Z. Li, X. Gao, Z. Y. Chen, Y. Duan, and F. Yang. Prompt-free efficient adaptation of segment anything model for remote sensing landslide detection. *IEEE Trans. Circuits Syst. Video Technol.*, 36(7):10564–10576, 2026. https://doi.org/10.1109/TCSVT.2026.3686175.

M. Liao, H. Wen, and L. Yang. Identifying the essential conditioning factors of landslide susceptibility models under different grid resolutions using hybrid machine learning: a case of Wushan and Wuxi counties, China. *Catena*, 217:106428, 2022. https://doi.org/10.1016/j.catena.2022.106428.

P. Lima, S. Steger, and T. Glade. Counteracting flawed landslide data in statistically based landslide susceptibility modelling for very large areas: a national-scale assessment for Austria. *Landslides*, 18(11):3531–3546, 2021. https://doi.org/10.1007/s10346-021-01754-y.

S. Liu, Z. Zeng, and T. Ren. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In *Proc. Eur. Conf. Comput. Vis. (ECCV)*, pages 38–55, 2024. https://doi.org/10.1007/978-3-031-72970-6_3.

T. Liu, T. Chen, and R. Niu. Landslide detection mapping employing CNN, ResNet, and DenseNet in the Three Gorges Reservoir, China. *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, 14:11417–11428, 2021. https://doi.org/10.1109/JSTARS.2021.3117975.

Z. Liu, M. P. Kwan, J. Wu, and H. Li. GeoPhysAdapter: Scale-matched geophysical adaptation for cross-domain landslide mapping with vision foundation models. *ISPRS J. Photogramm. Remote Sens.*, 243:140–159, 2026. https://doi.org/10.1016/j.isprsjprs.2026.09.035.

M. Loche, M. Alvioli, I. Marchesini, H. Bakka, and L. Lombardo. Landslide susceptibility maps of Italy: lesson learnt from dealing with multiple landslide types and the uneven spatial distribution of the national inventory. *Earth-Science Reviews*, 232:104125, 2022.

J. Luo, T. Luo, and M. Wang. Learning domain generalized remote sensing image segmentation by multiscale instance disentanglement. *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, 18:13538–13553, 2025. https://doi.org/10.1109/JSTARS.2025.3568853.

C. V. Nguyen, T. Hassner, M. Seeger, and C. Archambeau. LEEP: A new measure to evaluate transferability of learned representations. In *Proc. Int. Conf. Mach. Learn. (ICML)*, pages 7294–7305, 2020.

U. Ozturk, M. Pittore, R. Behling, S. Roessner, L. Andreani, and O. Korup. How robust are landslide susceptibility estimates? *Landslides*, 18(2):681–695, 2021. https://doi.org/10.1007/s10346-020-01485-5.

M. Pándy, A. Agostinelli, J. Uijlings, V. Ferrari, and T. Mensink. Transferability estimation using Bhattacharyya class separability. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pages 9162–9172, 2022. https://doi.org/10.1109/CVPR52688.2022.00896.

J. Peng, Y. Huang, and W. Sun. Domain adaptation in remote sensing image classification: A survey. *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, 15:9842–9859, 2022. https://doi.org/10.1109/JSTARS.2022.3220875.

S. Ren, K. He, R. Girshick, and J. Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. *IEEE Trans. Pattern Anal. Mach. Intell.*, 39(6):1137–1149, 2017. https://doi.org/10.1109/TPAMI.2016.2577031.

Q. Tang, C. Liu, and S. Zhang. Prompt-free universal region proposal network. *arXiv preprint arXiv:2603.17554*, 2026.

S. L. Ullo, A. Mohan, and A. Sebastianelli. A new Mask R-CNN-based method for improved landslide detection. *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, 14:3799–3810, 2021. https://doi.org/10.1109/JSTARS.2021.3064981.

G. Vivone, M. Dalla Mura, and A. Garzelli. A benchmarking protocol for pansharpening:

Dataset, preprocessing, and quality assessment. *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, 14:6102–6118, 2021. https://doi.org/10.1109/JSTARS.2021.3086877.

Yong Wei, Xiaodi Fu, Bo Zhang, Xiaohui Qin, Pinglang Kou, Linjun Wang, and Huajin Li. Efficient multi-source deep learning for rapid landslide mapping in the Karst mountains of Bijie, China. *Geomatics, Natural Hazards and Risk*, 17(1):2608246, 2026. https://doi.org/10.1080/19475705.2025.2608246.

Y. Xu, C. Ouyang, and Q. Xu. CAS landslide dataset: A large-scale and multisensor dataset for deep learning-based landslide detection. *Sci. Data*, 11(1), 2024. https://doi.org/10.1038/s41597-023-02847-z.

Yongpeng Yang, Jingwei Chen, Ya Guo, Xin He, Lei Wang, Chao Teng, Yu Bian, Zhixuan Zhang, and Meng Dai. The effect of hybrid models integration sequence: taking Fushun area as an example. *Geomatics, Natural Hazards and Risk*, 17(1):2707855, 2026. https://doi.org/10.1080/19475705.2026.2707855.

K. You, Y. Liu, M. Long, and J. Wang. LogME: Practical assessment of pre-trained models for transfer learning. In *Proc. Int. Conf. Mach. Learn. (ICML)*, pages 12133–12143, 2021.

Xia Zhao, Wei Chen, Paraskevas Tsangaratos, and Ioanna Ilia. Evaluating landslide susceptibility: the impact of resolution and hybrid integration approaches. *Geomatics, Natural Hazards and Risk*, 15(1):2409198, 2024. https://doi.org/10.1080/19475705.2024.2409198.